

Lesson 31: The Binomial Distribution — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. State the two parameters of $B(n, p)$.
2. Mean of $B(10, 0.5)$?
3. Mean of $B(20, 0.3)$?
4. Mean of $B(40, 0.25)$?
5. Which GDC function gives $P(X = r)$?
6. Which GDC function gives $P(X \leq r)$?

Section 2 — Core practice

7. $X \sim B(8, 0.5)$. Find $E(X)$.
8. $X \sim B(50, 0.2)$. Find $E(X)$.
9. $X \sim B(100, 0.1)$. Find the mean.
10. $X \sim B(20, 0.5)$. Find the variance $np(1-p)$.
11. Write $P(X \geq 1)$ in terms of $P(X = 0)$.

12. $X \sim B(12, 0.25)$. Find $E(X)$.

Section 3 — Challenge & exam-style

13. A quiz has 12 questions, $P(\text{correct guess}) = 0.25$, $X \sim B(12, 0.25)$. Find $E(X)$ and $P(X = 3)$.

14. A fair coin is tossed 6 times. Find $P(\text{exactly 4 heads})$.

15. $X \sim B(10, 0.4)$. Find $P(X \leq 3)$ (use Cdf, 3 s.f.).

16. $X \sim B(5, 0.2)$. Find $P(X \geq 1)$.